Econometrics

Working with Probability

Outilne

- What is probability?
- What kind of the probabilities do we have?
  - Marginal probability
  - Conditional probability
  - The law of iterated expectation
- Covariance and Correlation

Random Experiment and Outcome

…a ***random experiment*** is an action or process that leads to one of several ***possible outcome***s. For example:

Experiment	Outcomes
Flip a coin	Head, Tail
Flip a coin twice	(H,H), (H,T), (T,H), (T,T)
Exam Marks	Numbers: 0, 1, 2, ..., 100
Assembly Time	$t > 0$ seconds
Course Grades	F, D, C, B, A, A+

Sample Space

- A set of ***collectively exhaustive*** and ***mutually exclusive*** outcomes is called a ***sample space*** and is denoted by *S*.

Collectively Exhaustive: ALL possible outcomes included.
Mutually exclusive: no two outcomes can occur at the same time.

- The outcomes are denoted by $O_1, O_2, \ldots, O_k$
- We can represent the sample space and its outcomes as:

  $S = \{O_1, O_2, \ldots, O_k\}$

Event

- ***Event***: An event, say A, is a subset of the sample space and describes a possible type of occurrence.
  that is, an event is a set of one or more outcomes

  e.g. A={H}, B={H,T};
  A={1,3,5}, B={3,4,5,6};
  A={cash, debit}, B={debit, credit card, check}

Probability

- Given a sample space $S = \{O_1, O_2, \ldots, O_k\}$, the ***probability assigned to an event*** is a value between zero and one, inclusive, describing the relative possibility (chance or likelihood) the event will occur.
  e.g. Flip a coin, P({head}) = P({tail}) = 0.5.

- The sum of the probabilities of all the outcomes equals 1:

$$\sum_{i=1}^{k} P(\{O_i\}) = 1$$

e.g. P({head}) + P({tail}) = 1

Random Variable

- When the numerical value that a variable assumes presents the result of a random occurrence, that variable is a **Random Variable**.

Discrete Random Variable
– takes on ***finite*** or ***infinite but countable*** number of different values
– possible to list all results and the associated probabilities.

Continuous Random Variable
– takes on any value in an interval.
– cannot associate probability with a single value, only a range of values.

Probability Distribution

- A ***probability distribution*** is a table, formula, or graph that describes the values of a random variable and the probability associated with these values.

- Since a **random variable** can be discrete or continuous, we have two types of probability distributions:
  - Discrete Probability Distribution
  - Continuous Probability Distribution

- (Discrete) Probability ***mass*** function $f(x)$, which equals to $P(X = x)$, specifies the ***probability*** of each discrete outcome.

$$0 \leq f(x) \leq 1, \text{ and } \sum f(x_i) = 1$$

- (Continuous) Probability ***density*** function $f(x)$: a continuous function that describes the ***relative likelihood*** on a given value.

$$f(x) \geq 0, \text{ and } \int f(x)\, dx = 1$$

- Notice that for continuous random variable, we have $P(X = x) = 0, for\ any\ x \in R$.

- ***Cumulative*** Distribution Function $F(x)$, which equals to $P(X \leq x)$, specifies the probability that the random variable X will be less or equal to x.

- For discrete random variable,

$$P(X \leq x) = \sum_{u \leq x} f(u)$$

For continuous random variable,

$$P(X \leq x) = \int_{-\infty}^{x} f(u)du$$

Expected Value and Variance

	Continuous Random Variable	*Discrete Random Variable*
EV	$E(x) = \mu = \int_{-\infty}^{+\infty} x f(x) dx$	$E(x) = \mu = \sum_{all\ x} xP(x)$
Variance	$V(x) = \sigma^2 = \int_{-\infty}^{+\infty} (x-\mu)^2 f(x) dx$	$V(x) = \sigma^2 = \sum_{all\ x} [x-\mu]^2 P(x)$
EV↔Var	$V(x) = E[(X - E(X))^2]$ Note that $E(X)$ is a constant!	

Laws of Expected Value

1. $E(c) = c$

   The expected value of a constant (c) is just the value of the constant.

2. $E(X + c) = E(X) + c$

3. $E(cX) = cE(X)$

   We can "pull" a constant out of the expected value expression (either as part of a sum with a random variable X or as a coefficient of random variable X).

Laws of Variance

1. $\text{Var}(c) = 0$

   The variance of a constant (c) is zero.

2. $\text{Var}(X + c) = \text{Var}(X)$

   The variance of a random variable and a constant is just the variance of the random variable.

3. $\text{Var}(cX) = c^2\text{Var}(X)$

   The variance of a random variable and a constant coefficient is the coefficient squared times the variance of the random variable.

Exp1. Expectation

- Person A and Person B produce product X and product Y, respectively, with the corresponding defective rates as follows:

X	0	1	2	3
Prob	0.3	0.3	0.2	0.2

Y	0	1	2	3
Prob	0.2	0.5	0.3	0.0

Exp1. Expectation

- Find out whose defective rate is higher?

X	0	1	2	3
Prob	0.3	0.3	0.2	0.2

Y	0	1	2	3
Prob	0.2	0.5	0.3	0.0

Exp1. Expectation

- Since E(X) = 1.3; E(Y) = 1.1

Ans: Person A has higher defective rate.

X	0	1	2	3
Prob	0.3	0.3	0.2	0.2

Y	0	1	2	3
Prob	0.2	0.5	0.3	0.0

Exp2. Expectation

- For a random variable X with probability density function $f(x)$, calculate its expected value:

$$f(x) = \begin{cases} 2x, 0 < x < 1 \\ 0, otherwise \end{cases}$$

Conditional Distribution

- The distribution of a random variable Y conditional on another random variable X taking on a specific value is called the **conditional distribution** of Y given X.

- The conditional probability that Y takes on the value y when X takes on the value x is written $P(Y = y|X = x)$.

Conditional Distribution

- In general, the conditional distribution of Y given $X = x$ is

$$P(Y = y|X = x) = \frac{P(Y = y, X = x)}{P(X = x)}$$

Conditional Expectation

- The **conditional expectation** of Y given X, is the mean of the conditional distribution of Y given X.
- If Y takes on k (discrete) values $y_1, \cdots, y_k$, then the conditional mean of Y given $X = x$ is:

$$E[Y|X] = \sum_{i=1}^{k} y_i \, P(Y = y_i | X = x)$$

The Law of Iterated Expectations

- The expectation of Y is the expectation of the conditional expectation of Y given X,

$$E(Y) = E[E(Y|X)]$$

– where the inner expectation on the right-hand side of equation is computed using the conditional distribution of Y given X and the outer expectation is computed using the marginal distribution of X.

The Law of Iterated Expectations

- For a set of trivariate random variables, X, Y, and Z:

$$E(Y|X,Z) = \text{conditional mean given } X \text{ and } Z$$

$$E(Y|Z) = E[E(Y|X,Z)|Z]$$

$$E(Y|X) = E[E(Y|X,Z)|X]$$

Exp. LIE

Let X=schooling of the person
Y=monthly income of the person
Z=gender (Z=1 men; Z=2 women)

a. Check if there is salary discrimination on women

b. Solve the expectation of Y from the following table

X	P(X)	Z	P(Z)	Conditional Income Expectation
1	0.5	1	0.5	E(Y|X=1,Z=1)=1200
		2	0.5	E(Y|X=1,Z=2)= 800
2	0.5	1	0.5	E(Y|X=2,Z=1)=400
		2	0.5	E(Y|X=2,Z=2)=500

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Exp. LIE

Solution: Step 1:average out X

a. When Z=1,

X	P(X)	Conditional Income Expectation
1	0.5	E(Y|X=1,Z=1)=1200
2	0.5	E(Y|X=2,Z=1)=400

Use the formula above, **average out X**:

let: $E(Y|X=xi,Z=1)=a1$

$E(A|Z=1)= \sum_x E(Y|X=xi,Z=1)*P(X=xi)$

$= E[E(Y|X=xi,Z=1)|Z=1]$

Exp. LIE

Similarly,

when Z=2,

X	P(X)	Conditional Income Expectation
1	0.5	E(Y|X=1,Z=2)=800
2	0.5	E(Y|X=2,Z=2)=500

average out X:

$E(A|Z=2) = \sum_x E(Y|X=x_i,Z=2)*P(X=x_i)$

Exp. LIE

We can get that $E(A|Z=1) = 800$, and $E(A|Z=2) = 650$.

As $650 < 800$, the conditional expectations suggest that there is salary discrimination on women.

Independence

- Two random variables X and Y are **independently distributed**, or **independent**, if knowing the value of one of the variables provides no information about the other.
- Specifically, X and Y are independent if the conditional distribution of Y given X equals the marginal distribution of Y. That is,

$$P(Y = y|X = x) = P(Y = y)$$

Independence

- We have known that:

$$P(Y = y|X = x) = \frac{P(Y = y, X = x)}{P(X = x)}$$

- So, if X and Y are independent, then

$$P(Y = y|X = x) = P(Y = y)P(X = x)$$

Covariance and Correlation

- One measure of the extent (linear pattern) to which two random variables move together is their covariance.

- Formally, the **covariance** between X and Y is:

$$Cov(X, Y) = \sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)]$$

Covariance between two random variables

- Continuous random variables:

$$Cov(X,Y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x - \mu_X)(y - \mu_Y)\, f(x,y)dxdy$$

- Discrete random variables:

$$Cov(X,Y) = \sum_{all(x,y)}(x - \mu_X)(y - \mu_Y)\, P(x,y)$$

Laws of Covariance

1. $\mathrm{Cov}(X, Y) = E(XY) - E(X)E(Y)$
2. $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$
3. $\mathrm{Cov}(X, C) = 0$, where C is a constant
4. $\mathrm{Cov}(aX+b, cY+d) = ac\mathrm{Cov}(X, Y)$
5. $\mathrm{Cov}(X+Y, Z) = \mathrm{Cov}(X, Z) + \mathrm{Cov}(Y, Z)$

Correlation

- Because the covariance is the product of X and Y, deviated from their means, its units are, awkwardly, the units of X multiplied by the units of Y.
- This "units" problem can make numerical values of the covariance difficult to interpret.
- The correlation is an alternative measure of dependence between X and Y that solves the "units" problem of the covariance.

Correlation

- This “units” problem can make numerical values of the covariance difficult to interpret.

- For example, the covariance between the ages and salaries of CEO’s is 248.32 dollar-year. On the other hand, the covariance between the incomes and heights of male CEO’s is 77.72 dollar-inch.
- Which pair has stronger relationship?

Correlation

- This "units" problem can make numerical values of the covariance difficult to interpret.

- The **correlation** is an alternative measure of dependence between X and Y that solves the "units" problem of the covariance.

Correlation

- Specifically, the **correlation coefficient** between X and Y is the covariance between X and Y divided by their standard deviations:

$$Corr(X,Y) = \rho = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}}$$

- Because the units of the numerator in the equation are the same as those of the denominator, the units cancel, and the correlation is unit free.

Interpretation of Correlation

- Correlation coefficient suggests that one may have a statistical relationship between two variables, the change of one variable is associated in certain ***linear*** fashion with the change of another variable.

- Moreover, the correlation is always between -1 and 1:

$$-1 \leq Corr(X, Y) \leq 1$$

Interpretation of Correlation

- The correlation is always between -1 and 1:

$$-1 \leq Corr(X, Y) \leq 1$$

- When $\rho = 1$ or $\rho = -1$, we have "*perfect*" correlation.
  - A deterministic linear relationship is one in which knowledge of the value of one variable accompany the value of the other *exactly*.
  - $Y = aX + b$, for $a \neq 0$. If $a > 0, \rho = 1$, if $a < 0, \rho = -1$.

Interpretation of Correlation

- The correlation is always between -1 and 1:

$$-1 \leq Corr(X, Y) \leq 1$$

- Correlation coefficient measures strength and direction of *linear* relationships only.

✓ $\rho = 0$ generally implies that no linear relationship.

✓ $\rho > 0$ means both variables increase at the same time.

✓ $\rho < 0$ means one variable increases as the other decreases.

Interpretation of Correlation

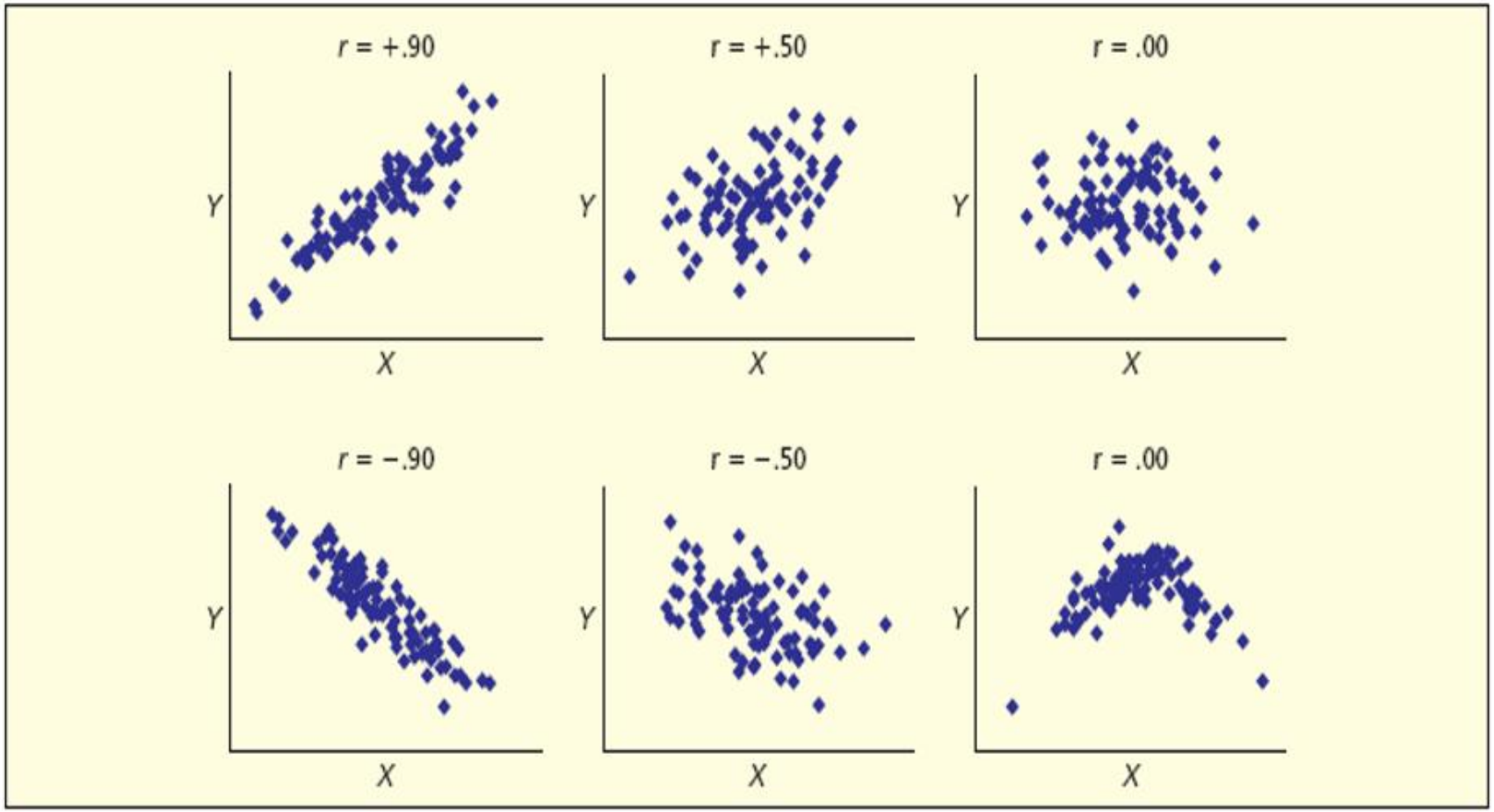
- The correlation is always between -1 and 1:

$$-1 \leq Corr(X, Y) \leq 1$$

- Finally, we say that two random variables X and Y are **uncorrelated** if

$$Cov(X, Y) = 0$$

Illustration of Correlation Coefficients



Causality vs. Correlation

- Correlation coefficient tells us **nothing** about "causality".

– 'Spurious' correlation interprets a large ρ as suggesting a causal relationship b/w two variables that doesn't exist.
– In many cases, correlation is just describing that the two move together by luck or due to another variable.

Causality vs. Correlation

- Correlation coefficient tells us **nothing** about "causality".

- In fact, this is an important **distinction between econometrics and statistics**. For econometricians, they aim to uncover the causal effect of one variable on another variable.

Uncorrelated and Independent

- Two random variables X and Y are uncorrelated, means that there is **no linear relationship** between them.
- Two random variables X and Y are independent, means that there is **no relationship** between them.
- Q: which one is stronger?

Uncorrelated and Independent

- Two random variables X and Y are uncorrelated, means that there is **no linear relationship** between them.
- Two random variables X and Y are independent, means that there is **no relationship** between them.
- **Ans**: Independence excludes all forms of association between random variables, while uncorrelation only rules out linear associations!

Combination of more than one variables

- More than one RV can be linearly combined to create a new random variable ($Z = X + Y$).

Q: How can we calculate the expected value and variance of the new random variable?

More than One RV: Operator E and V

Expectation Operator *E* and Variance Operator *V*

Operator E	Operator V
$E[a X + b Y] = a E[X] + b E[Y]$	$V[a X + b Y] = a^2 V[X] + b^2 V[Y] + 2ab\ cov[X,Y]$ (if X, Y correlated) $= a^2 V[X] + b^2 V[Y]$ (if X, Y uncorrelated*)
$E[a X + b Y + c Z] = a E[X] + b E[Y] + c E[Z]$	$V[a X + b Y + c Z] = a^2 V[X] + b^2 V[Y] + c^2 V[Z] + 2ab\ cov[X,Y] + 2bc\ cov[Y,Z] + 2ca\ cov[Z,X]$ (if all pairs of X,Y, Z correlated)

Conclusion

- Any random variable has its own achievement probability and distribution.
- Conditional expectation is important for relating one variable to another variable.
- The law of iterated expectation is powerful!
- Correlation is not equal to causality in principle.

- Next time: Sampling Distribution and CLT